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IMAGING INFRARED SEEKER DESIGN

A THESIS SUBMITTED

TO

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ABSTRACT

The subject of this study is the design of an imaging infrared seeker for aerial targets. Firstly the radiant power of the target was discussed and optical design of seeker lenses were done with ZEMAX Optical Design Software considering the MTF (Modulation Transfer Function) values, optical path differences, distortions and detection range calculations. After that, detection range calculations and simulation results of the optical system were presented. Before lock on range calculations, spectral band of the system was analyzed and chosen for aerial targets. After t

hat, lock on range was calculated considering detector noise, background noise and target radiation. Afterward the optical parameters of the seeker were determined. Focal length and F# (F-number) of the seeker was calculated as it was receiving the aerial target and background radiation at 5 km distance. Design was done using ZEMAX optical design program. Moreover, the lens materials were Silicon, Germanium and Cadmium Selenide and Sapphire were dome material. Finally, optical performance parameters have been analyzed. Finally, test plate fitting is applied to the optical system. Manufacturing parameters were presented and discussed.

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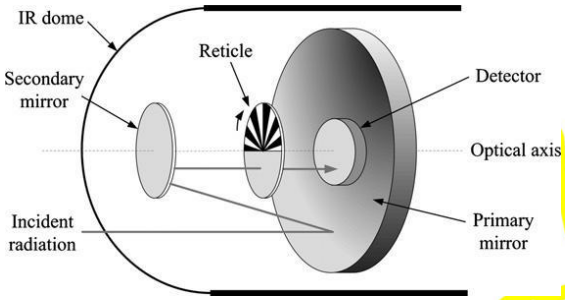
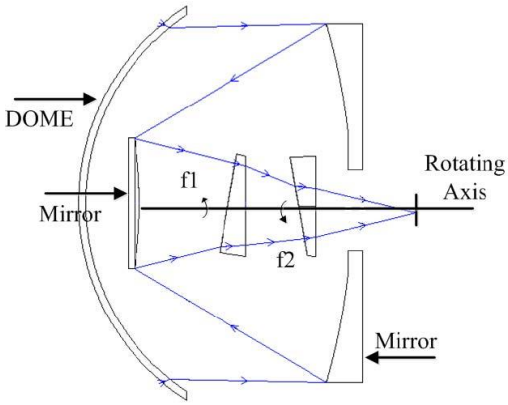
These types of seekers are also called spin-scan seekers. The optical structure of the spin-scan seeker is shown in the Fig. 3.4. There are four optical components, primary mirror, secondary mirror, reticle and detector. Primary and Secondary mirror is used to collect the radiation emitted from the target onto the detector and reticle is used to modulate (amplitude modulation) the infrared radiation. This type of modulation is also called Amplitude-Phase modulation.

Target signal and reference signal is compared while finding the angular position of target. When there is no target in the FOV of seeker the output of the detector is shown in Fig. 3.5 a). When there is a target in the FOV of the seeker, the output of the detector is shown is Fig. 3.5 b) and c). The difference in Fig. 3.5 b) and c) comes from the different angular positions of the target.



3.1.3 Third Generation Infrared Seekers Among the various IR seekers, the rosette scanning infrared seeker RSIS is a tracker in which a single detector scans the total field of view FOV in a rosette pattern. The rosette pattern is achieved by means of two counter-rotating optical elements such as prisms, tilted mirrors, or off centered lenses [13]. The optical structure of this type of missile is shown in Fig. 3.7. In this figure two wedges are counter-rotating. This provides the seeker rosette scanning property.